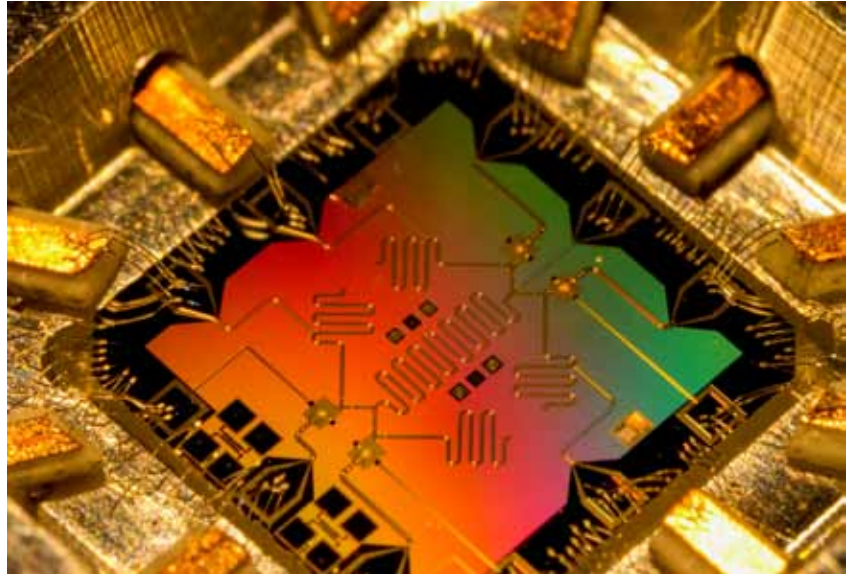


3. **Rigetti Computing**, a Californian firm, employs microwaves to control pairs of electrons flowing through superconducting circuits.
4. **Microsoft** is working on a ‘topological’ quantum computer that relies on the interactions of super-cold electrons.
5. **Psi Quantum** does its computing with photons that run along waveguides etched onto ordinary silicon chips. It hopes to leapfrog the noisy intermediate-scale quantum (NISQ) technology era entirely and produce a full-fledged quantum computer within the next five years. One of the firm’s founders says that Psi Quantum is already producing wafers full of quantum-computing chips in partnership with Global Foundries, a big contract chip-maker.
6. **Q-CTRL**, an Australian start-up, has raised tens of millions of dollars from investors including Sequoia Capital and In-Q-Tel. Their engineers have used machine learning to improve hardware-control routines written initially by human beings. This can cut error rates by 90 percent and reduce variability across an individual chip by a similar amount.
7. **IBM** recently committed to having a functioning 1,000-qubit quantum computer by 2023. With the IBM Quantum Condor planned for 2023, running 1121 qubits to be exact, quantum computers would start to tackle a substantial number of genuine real-world calculations, rather than being restricted to laboratory experiments. The director of IBM Research hopes to bring error rates down from one percent today to around 0.0001 percent in the Condor processor. It is a bold target considering IBM’s biggest quantum computer to date holds just 65 qubits. The company says it plans to have a 127-qubit machine ready in 2021, a 433-qubit one available in 2022, and a computer holding a million qubits and hopefully a functioning 1000-qubit quantum computer by



Model of the quantum processor computer

2023. One thousand qubits in total still would not be enough to take on full-scale quantum computing challenges, but it would be enough to maintain a small number of stable, logical qubit systems that could then interact with each other.

8. **Xanadu** quantum computers based on photons may possess key advantages over those based on electrons. The Canadian quantum computing start-up, Xanadu (a part of IBM’s quantum computing Q-network), has, for the first time, made a photonic quantum computer publicly available over the cloud.

The classic approach to photonic quantum computing, a linear optical quantum computing, relies on qubits, each based on a single photon. This strategy manipulates photons with mirrors, beam splitters, and phase shifters. Single photon detectors are then used to help read the results of what these devices have done. The problem with this approach is that single photons are difficult to experiment with, generally limiting this strategy to a handful of photons. In contrast, Xanadu’s strategy, known as Continuous Variable Quantum Computing, does not employ single-photon generators. Instead, the company relies on so-called ‘squeezed

states’ consisting of superpositions of multiple photons.

Some new technologies that are being explored in the field of quantum computing are described below.

Photonics: If a stray photon, which is a particle of light, from outside the system were to interact with a qubit, its wave would interfere with the qubit’s superposition, essentially turning the calculations into a jumbled mess—a process called decoherence. It turns out that superconducting cavities made of niobium, normally used to propel particle beams in accelerators, could be the solution. These cavities need to be constructed very precisely and operate at very low temperatures to efficiently propagate the radio waves that accelerate particle beams. Researchers, for instance at Fermilab and Argonne, theorise that by placing quantum processors in these cavities, the qubits will be able to interact undisturbed for seconds rather than the current record of milliseconds, giving them enough time to perform complex calculations.

The photons are encoded using thermo-optical phase shifters that are controlled by electrical voltages. Different settings of the phase shifters control the photon’s transmission